Paper 2, Paper 4 Exam Paper (With Mark Schemes)

Candidate Name: ____

Time Allowed: 57 minutes

Total Marks: 48

Instructions:

• Answer all questions.

• Click the blue '■ View Mark Scheme' link to jump to answers.

Question 1

(a) Simplify each of the following expressions, leaving your answers in index form.

(i) $3^4 \times 3^2$

.... [1]

(ii) $\frac{5^7}{5^4}$

.... [1]

(iii) $(2^3)^4$

.... [1]

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Question 2

(a) Evaluate each of the following numerical expressions. Give your answers as integers or fractions in their simplest form.

(i) 7^0

.... [1]

(ii) 4^{-2}

.... [1]

(iii) 3×2^{-3}

.... [1]

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Question 3

(a) Find the exact value of each expression.

(i) $25^{\frac{1}{2}}$

.... [1]

(ii) $27^{\frac{1}{3}}$

.... [1]

(iii) $16^{\frac{3}{4}}$

.... [2]

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Question 4

(a) Simplify the following operations completely.

(i) $2^{-4} \times 2^6$

.... [1]

(ii) $\dfrac{6^2}{6^{-1}}$

.... [1]

(iii) Show that the expression $\left(\dfrac{2}{3}\right)^{-2}$ reduces exactly to $2\dfrac{1}{4}$.

[2]

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Question 5

(a) Work out the value of each expression, giving your answer as a single integer or a simplified fraction.

(i) $8^{-\frac{1}{3}}$

.... [2]

(ii) $\left(\dfrac{9}{4}\right)^{-\frac{1}{2}}$

.... [2]

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Question 6

(a) Evaluate each of the following expressions completely, giving your final answer as an exact integer or a fraction in its simplest form.

(i) $\left(\frac{1}{8}\right)^{-\frac{2}{3}}$

.... [2]

(ii) $125^{\frac{1}{3}} \times 2^{-3}$

.... [2]

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Question 7

(a) Solve the following index equations for x.

(i) $3^x = \frac{1}{81}$

.... [2]

(ii) $4^{2x-1} = 32$

.... [2]

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Question 8

(a) Simplify the operational expressions.

(i) Find the exact value of the expression.

$$\frac{2^5 \times 6^3}{3^2 \times 4^3}$$

.... [3]

(ii) Show that the expression reduces exactly to an integer.

$$\frac{\left(5^{\frac{1}{2}}\right)^4 \times 10^2}{2^2 \times 5^3}$$

[2]

■ View Mark Scheme

Question 9

(a) Work out the value of the following composite expressions.

(i) $\left(\frac{16}{81}\right)^{-\frac{3}{4}}$

.... [2]

(ii) $\left(4^{\frac{3}{2}} + 27^{\frac{2}{3}}\right)^{-1}$

.... [3]

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Question 10

(a) Evaluate each of the operations without using a calculator, leaving answers as standard rational values.

(i) $\left(2\frac{7}{9}\right)^{-\frac{1}{2}}$

.... [2]

(ii) $\dfrac{3^{-5} \times 9^{2}}{27^{-1}}$

.... [2]

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Question 11

Solve the simultaneous index equations for x and y.

$$2^{x} \times 4^{y} = \frac{1}{8}$$

$$27^{x} \times 3^{y} = 9^{\frac{1}{2}}$$

.... [4]

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Question 12

(a) An advanced engineering formula determines the operational efficiency metric, E, based on an input index variable x such that:

$$E = \frac{243^{\frac{x}{5}} \times 4^{-x}}{3^{x-1}}$$

(i) Show analytically that the expression for E simplifies exactly to 3×4^{-x}.

[2]

(ii) Find the value of E when $x = 2.7$, correct to 3 significant figures.

.... [2]

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Topic Checklist & Self-Reflection

Review the topics below and circle your confidence level.

Question	Topic Details	Confidence
Question 1	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]
Question 2	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]
Question 3	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]
Question 4	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]
Question 5	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]
Question 6	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]
Question 7	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]
Question 8	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]
Question 9	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]
Question 10	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]
Question 11	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]
Question 12	**Arithmetic Indices and Power Rules**	[Low] [Med] [High]

Mark Schemes

Question 1 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	3^6	1	B1
(b)	5^3	1	B1
(c)	2^{12}	1	B1

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Question 2 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	1	1	B1
(b)	$\frac{1}{16}$	1	B1
(c)	$\frac{3}{8}$	1	B1

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Question 3 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	5	1	B1
(b)	3	1	B1
(c)	8	2	B2 for correct answer, or M1 for $(\sqrt[4]{16})^3$ or 2^3 or $\sqrt[4]{4096}$

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Question 4 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	2^2 or 4	1	B1
(b)	6^3 or 216	1	B1
(c)	$2\frac{1}{4}$	2	M1 for $\left(\frac{3}{2}\right)^2$ or $\frac{1}{\left(\frac{2}{3}\right)^2}$ or $\frac{2^{-2}}{3^{-2}}$ or $\frac{9}{4}$

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Question 5 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	$\frac{1}{2}$	2	B2 for correct answer, or M1 for $\frac{1}{8^{\frac{1}{3}}}$ or $\frac{1}{2}$ seen in working
(b)	$\frac{2}{3}$	2	B2 for correct answer, or M1 for $\left(\frac{4}{9}\right)^{\frac{1}{2}}$ or $\frac{1}{\sqrt{\frac{9}{4}}}$ or $\frac{3}{2}$ seen

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Question 6 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	4	2	M1 for $8^{\frac{2}{3}}$ or $(\sqrt[3]{8})^2$ or 2^2 or $\left(\frac{1}{4}\right)^{-1}$
(b)	$\frac{5}{8}$	2	M1 for $125^{\frac{1}{3}} = 5$ or $2^{-3} = \frac{1}{8}$ or $\frac{5}{2^3}$

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Question 7 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	-4	2	M1 for $3^x = 3^{-4}$ or $\frac{1}{81} = \frac{1}{3^4}$
(b)	$\frac{7}{4}$	2	M1 for $(2^2)^{2x-1} = 2^5$ or $2(2x-1) = 5$ or $4x - 2 = 5$

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Question 8 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	12	3	M2 for $\frac{2^5 \times 2^3 \times 3^3}{3^2 \times 2^6}$ or M1 for breaking down bases into prime factors: $6^3 = 2^3 \times 3^3$ or $4^3 = 2^6$
(b)	1	2	M1 for $\left(5^{\frac{1}{2}}\right)^4 = 5^2$ and $10^2 = 2^2 \times 5^2$ leading to $\frac{5^2 \times 2^2 \times 5^2}{2^2 \times 5^3}$ or $\frac{25 \times 100}{4 \times 125}$

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Question 9 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	$\frac{27}{8}$	2	M1 for $\left(\frac{81}{16}\right)^{\frac{3}{4}}$ or $\left(\frac{3}{2}\right)^3$ or $\frac{2^{-3}}{3^{-3}}$
(b)	$\frac{1}{17}$	3	M1 for $4^{\frac{3}{2}} = 8$, M1 for $27^{\frac{2}{3}} = 9$, then leading to $(8+9)^{-1}$

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Question 10 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	$\frac{3}{5}$	2	M1 for converting to improper fraction $\left(\frac{25}{9}\right)^{-\frac{1}{2}}$ or finding $\sqrt{\frac{9}{25}}$
(b)	4	2	M1 for writing all terms with base 3: $\frac{3^{-5}\times(3^2)^2}{(3^3)^{-1}}$ or $3^{-5+4-(-3)}$ or 3^2

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Question 11 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
x, y	$x=1, y=-2$	4	M1 for $x+2y=-3$, M1 for $3x+y=1$, M1 for attempting to solve the simultaneous linear system, A1 for both values correct.

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Question 12 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	3×4^{-x}	2	M1 for $243^{\frac{x}{5}}=(3^5)^{\frac{x}{5}}=3^x$ leading to $\frac{3^x\times 4^{-x}}{3^{x-1}}$ or $3^{x-(x-1)}\times 4^{-x}$
(b)	0.0712	2	M1 for $3\times 4^{-2.7}$ or $3\times 0.0237...$ or $0.07119...$ seen

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